

Electrolytic corrosion. Usually, corrosion requires presence of moisture and soluble impurities, which may be inherent in the materials or contained in atmospheric pollution. These provide the electrolyte necessary for electro-chemical reaction of the corrosion process. The reaction takes place where dissimilar metals are in contact or may also be initiated when an electrolyte bridges the gap between metal surfaces and current flows. It is not necessary to have visible wetting of the surface; an invisible adsorbed film of moisture is sufficient to support electrolytic corrosion.

The presence of conductive electrolyte on the surface of PCBs under voltage stress can result in surface tracking because of corrosion or, in the worst case, electro-migration of metal across the gap resulting in a short circuit.

The ingress of moisture into a connector may result in corrosion around pins. Corrosion in connectors can increase surface resistance with the possibility of over-heating culminating in loss of performance or even a fire.

Low humidity problems

Most of the low humidity effects are evident in hygroscopic materials. In some cases, moisture removal can change the mechanical structure and cause embrittlement. Often, this is also accompanied by shrinkage and weight loss.

Static damage. One major difficulty with static damage is that the results are not always immediately obvious because the resulting failure often destroys the evidence of damage. It is therefore difficult to categorically state that the failure was caused by static damage. The most critical period when damage can be inflicted is handling before a device is inserted into the board.

Dust. Static charge attracts dust. The buildup of dust in an unsealed unit can result in malfunction by caus-

ing bad contacts or creating a path that allows tracking of high-voltage discharges to the earth—such as in a TV or computer monitor.

Electrostatic generation. Static charge generation is greatly affected by humidity levels. Voltages as high as 20kV can be generated by a person walking across a carpet when humidity levels are below 30% RH. Under high humidity conditions, the same walk may generate only about 1.5kV. If this person picks up a device without following the correct antistatic measures, device damage can occur.

Static susceptibility. Non-catastrophic damage can be caused by a static discharge as low as 30V in some MOSFET technologies where the gate oxide is very thin. Junctions, gates and contacts are the critical areas within a device where electrostatic discharge damage can occur.

Protection measures

Organic coatings are extensively used to protect electronic assemblies from moisture, high humidity and high temperature environments. Some coatings have been specifically engineered and formulated to remove large amounts of heat when subjected to high tem-

peratures for short periods of time. Among these are the thermally ablative coatings that are used to protect components during missile and rocket liftoffs. These coatings, when applied to heat-sensitive parts such as electronic enclosures or nozzle control units, protect the inside surfaces from temperatures as high as 3000°F generated by the rocket engines during blast-off.

At the temperature of ablation, the exterior layers of coatings decompose and char. Thus the coating absorbs considerable heat (heat of ablation) while the inside surfaces remain sufficiently cool. The coating must be applied at a sufficient thickness and must ablate at a controlled rate, so that sufficient coating remains at the end of thermal exposure.

Ablative coatings consisting of filled epoxies and silicones were used to protect the nozzle control unit of Minuteman missile during take-off. A similar coating was used on Apollo capsule to protect astronauts from the intense heat of re-entry into the Earth's atmosphere. Intumescent coatings are used to protect electronic boxes and structures from fire. These contain fillers or molecular structures that decompose and evolve into flame-retardant gases such as carbon dioxide.

Among the best non-metallic, high-thermally conductive fillers are beryllia, aluminium nitride, boron nitride and diamond. At the same time, conformal coatings don't provide protection unless the surface is completely clean from flux residues, fingerprints as well as any other chemical used in the production of electronic assemblies.

Apart from temperature and humidity, electronic assemblies must be protected from marine environment (where salt spray/fog dominates), handling, abrasion, friction, radiation, micro-organism, etc. **EFY**

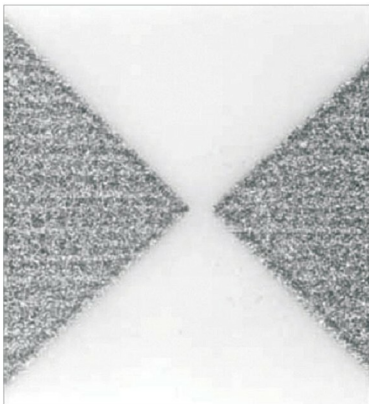


Fig. 3: Metal migration—water-drop test for thick-film gold conductor (no evidence of metal migration after 1 hour, 4V DC)